

NETWORK SECURITY

INTRODUCTION

In today's digital world, networks are the backbone of organizations, and protecting them from cyber threats is essential. Network Penetration Testing is a proactive approach used to identify vulnerabilities and weaknesses within network infrastructure before attackers can exploit them. By simulating real-world attacks, security professionals can assess how well the network is protected, uncover flaws, and implement corrective measures. Tools like **Nmap** and **Metasploit** are widely used to scan, exploit, and secure networks. This report explains the concept of network penetration testing, the tools used, steps involved, and how to protect networks against the vulnerabilities found during testing. It also includes the practical experience gained from completing the TryHackMe Metasploit Intro lab.

1. What is Network Penetration Testing?

Network Penetration Testing is a controlled and authorized process where testers simulate cyberattacks on a network to uncover security weaknesses. The goal is to find vulnerabilities such as open ports, misconfigured services, outdated software, and improper access controls before malicious attackers can exploit them. The findings are then used to strengthen defenses and ensure the network's reliability and safety.

2. What is Nmap, and why is it used?

Nmap (Network Mapper) is a powerful and open-source scanning tool used in network security. It helps testers identify live devices, discover open ports, detect services running on those ports, and fingerprint operating systems. Nmap assists penetration testers during the reconnaissance phase by mapping the network, identifying potential targets, and finding weaknesses that need to be secured.

3. What is Metasploit, and why is it used?

Metasploit is a framework that helps penetration testers execute exploits against vulnerable systems. It contains pre-built modules for attacks, payloads, and auxiliary tools. Metasploit allows users to test how attackers would exploit known vulnerabilities, gain access to systems, and move through networks. It is widely used for validating security defenses and improving network protection.

4. Basic Steps to Perform a Penetration Test using Nmap and Metasploit

Step 1 – Reconnaissance with Nmap

- Use Nmap to scan the target network for open ports and running services.
- Example command: `nmap -sS -A 192.168.x.x`

Step 2 – Identify Vulnerabilities

- Review the results to find outdated software, unsecured ports, or weak services that attackers could exploit.

Step 3 – Choose an Exploit in Metasploit

- Launch Metasploit with `msfconsole`.
- Search for exploits using known vulnerabilities: search `exploit_name`.

Step 4 – Configure and Launch the Exploit

- Set target IP, port, and payload options.
- Execute the exploit to see if the target system is vulnerable.

Step 5 – Post-Exploitation and Reporting

- If successful, access the target and gather information.
- Document the findings and recommend security improvements.

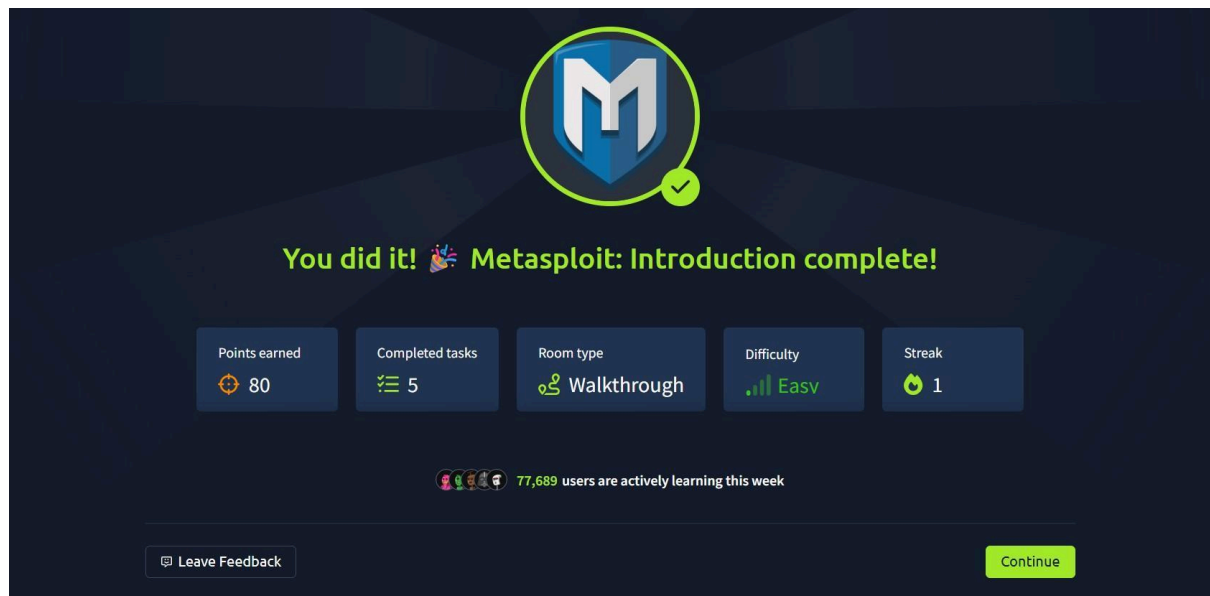
5. How to Protect a Network Against the Types of Vulnerabilities Found During Your Test

To ensure network safety, the following steps should be implemented:

1. **Apply Regular Patches and Updates** – Keep all software and devices updated with security patches.
2. **Use Firewalls and Access Control Lists (ACLs)** – Limit open ports and control access to sensitive resources.
3. **Enable Strong Authentication** – Enforce multi-factor authentication and password complexity rules.
4. **Disable Unnecessary Services** – Reduce the attack surface by turning off unused ports and services.
5. **Monitor Traffic and Logs** – Use intrusion detection systems to identify abnormal behavior.
6. **Educate Employees** – Provide training on secure practices like phishing prevention.
7. **Conduct Regular Security Audits** – Test networks frequently using tools like Nmap and Metasploit to identify vulnerabilities early.

6. Completion of TryHackMe Lab – Metasploit Introduction

I successfully completed the **Metasploit Intro Lab** on TryHackMe



This practical experience helped me understand how penetration testers operate and how tools like Metasploit assist in real-world security assessments.

CONCLUSION

Network Penetration Testing is a crucial part of cybersecurity, helping organizations proactively identify and mitigate risks. Tools like Nmap and Metasploit enable testers to simulate attacks, find weaknesses, and strengthen defenses. By applying input validation, network monitoring, authentication, and regular audits, organizations can protect themselves from malicious threats. Completing hands-on labs, such as the TryHackMe Metasploit Intro, reinforces the skills necessary to secure networks and respond to evolving security challenges.